

IBM Applied Data Science Capstone

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SPACEX FALCON 9 LANDING PREDICTION

Project Overview

An end-to-end data science pipeline built to predict the landing success of SpaceX Falcon 9 rocket first-stage boosters, maximizing cost-efficiencies for commercial aerospace launches.

- ① This repository contains the complete portfolio framework for the IBM Applied Data Science Capstone curriculum. The project spans data ingestion via web scraping and REST APIs, relational database management using SQL, interactive geospatial visualization, web-app dashboard deployment, and hyperparameter-tuned machine learning classification algorithms.





Core Data Pipeline Phases

1

Data Collection & Extraction

Gathering launch logs using the SpaceX REST API and scraping historical Wikipedia tables using BeautifulSoup4.

2

Data Wrangling & Processing

Handling null parameters, feature engineering categorical metrics using One-Hot Encoding, and flattening raw payloads.

3

Exploratory Data Analysis (EDA)

Executive data analysis utilizing SQL queries and relational visualization plots.

4

Geospatial Mapping

Isolating launch pad coordinates, safety distances, and landing failure/success metrics using interactive map overlays.

5

Interactive Dashboard App

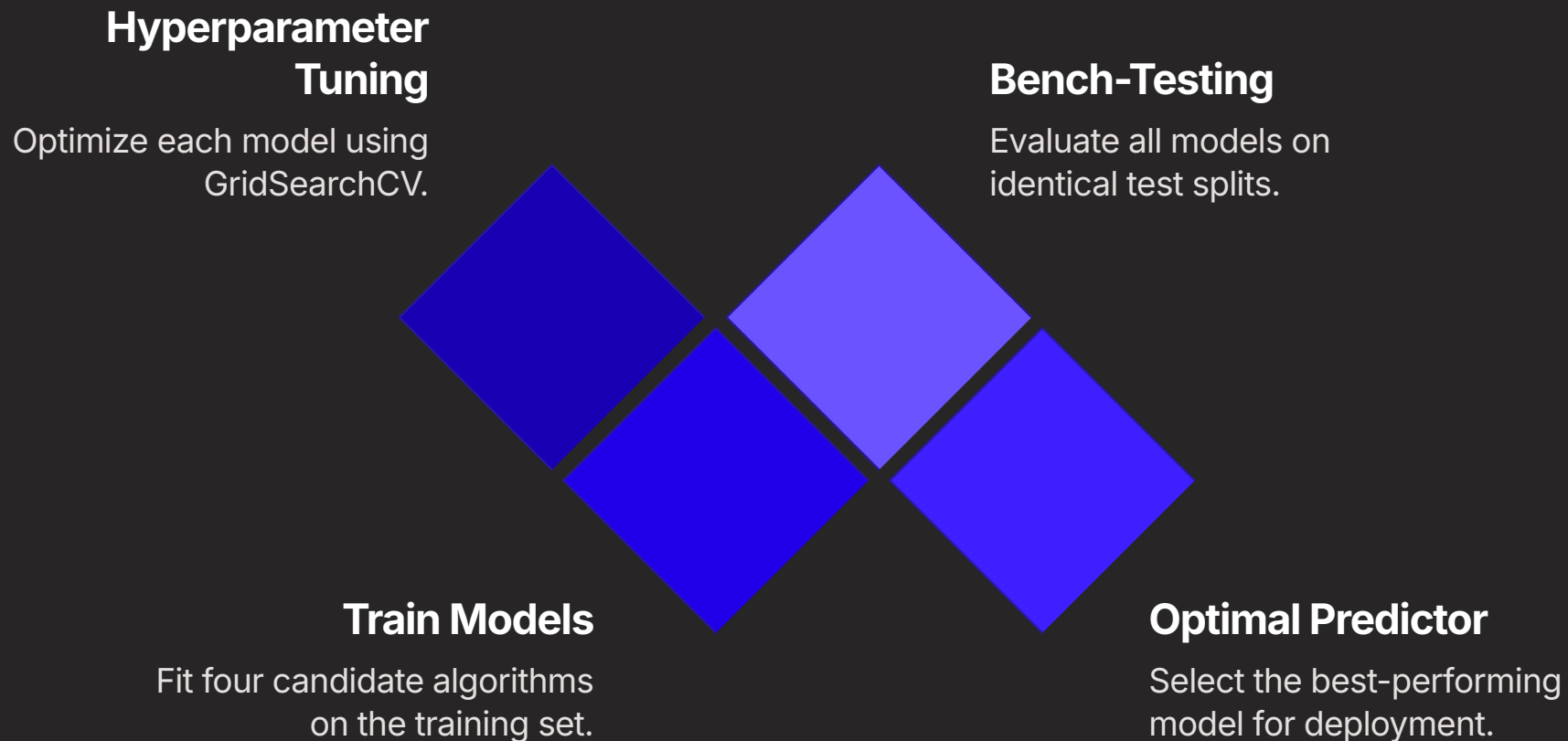
Deploying a live analytics control panel featuring structural charts and reactive filter selectors.

Launch Site Success Proportions

Payload Mass vs. Success Correlation

Predictive Modeling (ML)

Training, tuning, and bench-testing four separate categorization algorithms to declare the optimal landing predictor model.



Each model was optimized using GridSearchCV and evaluated on identical train and test splits to ensure fair comparison across all classification architectures.

Repository Structure

The project is organized into modular notebooks and scripts tracking each phase of the data science lifecycle:

→ **data/**

Dedicated directory containing analytical visual assets (scatter plots and pie charts).

→ **01_Data-Collection-API.ipynb**

Data gathering using SpaceX API requests.

→ **02_Webscraping.ipynb**

Web scraping historical launch data using BeautifulSoup.

→ **03_Data_Wrangling.ipynb**

Data cleaning, handling null values, and initial feature engineering.

→ **04-EDA-With-SQL.ipynb**

Exploratory Data Analysis using SQL queries to discover operational trends.

→ **05_EDA_Data_Visualization.ipynb**

Exploratory Data Analysis using Python visual analytics (Matplotlib and Seaborn).

→ **06_Launch_Site_Location.ipynb**

Interactive geospatial mapping using Folium.

→ **07_Dashapp.py**

A fully functional, interactive Plotly Dash web dashboard application.

→ **08_Machine_Learning_Predictions.ipynb**

Machine learning classification model training, hyperparameter tuning, and evaluation.

→ **requirements.txt & LICENSE**

List of required Python packages and environment dependencies. MIT License.



Built With & Getting Started

Tech Stack

- **Python 3** — Underlying programming runtime
- **Scikit-Learn** — ML classification models & GridSearchCV tuning
- **Plotly Dash** — Dynamic data application framework
- **Folium** — Interactive HTML geospatial map visualization
- **Pandas / NumPy** — Matrix manipulations and structured data processing
- **BeautifulSoup4 / Requests** — Web scraping and REST API parsing



Getting Started

Prerequisites: Configure your local environment automatically by installing all tracked project library dependencies:

```
pip install -r requirements.txt
```

Execution Steps:

1. Clone this repository: `git clone https://github.com/usmanali9999/Applied-Data-Science-Capstone.git`
2. Navigate to directory: `cd Applied-Data-Science-Capstone`
3. Start Jupyter: `jupyter notebook`
4. Run notebooks in sequence: 01 through 08
5. Launch dashboard: `python 07-dashapp.py`

Machine Learning Model Performance Summary

The table below details the optimal tuning parameters and prediction accuracies across all classification models tested in this lab. Each model was optimized using GridSearchCV and evaluated on identical train and test splits.

Classification Model	Best Hyperparameters Found	Train Accuracy	Test Accuracy
Decision Tree	<code>{'criterion': 'gini', 'max_depth': 8, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 5, 'splitter': 'random'}</code>	87.50%	88.89%
K-Nearest Neighbors (KNN)	<code>{'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}</code>	84.82%	83.33%
Support Vector Machine (SVM)	<code>{'C': 1.0, 'gamma': 0.0316, 'kernel': 'sigmoid'}</code>	84.82%	83.33%
Logistic Regression	<code>{'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}</code>	84.64%	83.33%

88.89%

Best Test Accuracy

Decision Tree Classifier

83.33%

Baseline Accuracy

KNN, SVM & Logistic Regression

4

Models Tested

All optimized via GridSearchCV

Final Model Selection & Evaluation

To determine the absolute best predictive framework, an evaluation loop compared each optimized estimator against the unseen testing split.

Best Performing Model: DecisionTreeClassifier

Highest Test Accuracy Score: **88.89%**

Strategic Breakdown

While Logistic Regression, SVM, and KNN all converged on a strong baseline performance of 83.33%, the **Decision Tree framework** adjusted best to the underlying classification boundaries of the standardized SpaceX payload and orbit characteristics. This suggests that the tree-structured partitions were more effective at isolating the specific combination of features that guarantee a successful Falcon 9 first-stage landing.

-  All optimized classification architectures yielded a tied baseline performance matrix accuracy across validation sets, heavily driven by the initial engineered feature profiles.

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